

Reflective Lantern - Innovation Report

Project

Project: Temporal-Pulse

URL: <https://github.com/atharvadevne123/reflective-lantern/tree/main/temporal-pulse>

Date: 2026-07-08

Commits: 60

Deliverables (15)

- FastAPI app with 9 endpoints: /detect /forecast /train /health /metrics /version /drift /feature-importance /anomalies
- Isolation Forest anomaly detector + Random Forest forecaster with 5-fold CV
- Multi-step forecasting with tree-level 90% confidence intervals
- FAISS nearest-neighbour root cause analysis (sklearn fallback)
- Feature pipeline: rolling stats, lags, rate-of-change, cyclical time encoding, cross-sensor correlations
- KS-test drift detection with reference/current distributions
- Sliding-window per-IP rate limiting middleware
- Structured JSON logging with correlation IDs
- SQLAlchemy models + Alembic initial migration (4 tables, composite indexes)
- Airflow-compatible daily retraining DAG
- Docker + docker-compose with PostgreSQL 16 and healthchecks
- GitHub Actions CI: ruff + pytest coverage + mypy (all green locally)
- Synthetic data generator with injected anomalies
- 152 tests passing; ruff and mypy clean
- Full README: quick start, API reference, architecture, config table

Test Status

PASS (152/152 tests, ruff clean, mypy clean)

Notes

- Session GitHub access is scoped to reflective-lantern only; new repo creation was blocked, so Temporal-Pulse ships as a subproject at temporal-pulse/ with its own CI workflow.
- Release tag push blocked by the git proxy; project is versioned v1.0.0 in pyproject.toml and CHANGELOG.